MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12SK7Y 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, rf remote cutoff

©	Outline	--- 8-1 (EIA)
	Base	--- B8-21 (low-loss phenolic)
	Envelope	--- MT8
	Cathode	--- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	sh	h	g3	g1	k	g2	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Pp	Pg2	© Ehk	Rg1	© Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	W	W	v	Meg	ft
Maximum:	13.8	330	---	140	---	4.4	0.44	100	---	(See
Minimum:	11.4	---	---	---	---	---	---	---	---	note 1)
TEST CONDITIONS (1):	12.6	250	-3	100	0	---	---	---	---	---
TEST CONDITIONS (2):	12.6	28	0	28	0	---	---	---	2	---

GENERAL:

Qualification - Required

1/ See note 4

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Test condition (1) (see note 2)	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (1) (anode)	Test condition (1)	0.65	II	Ib	6.5	12.0	mAdc
1306	Transconductance (1)	Test condition (1)	0.65	II	Sm	1,600	2,400	µmhos
1231	Emission	Eb = Ec1 = Ec2 = Ec3 = 30 Vdc (see note 2)	0.65	II	Is	65	---	mAdc
Ⓒ 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Test condition (1); Rp = 2,000 ohms	---	---	Ep	---	100	mVac
1301	Heater current		---	---	If	138	162	mA
Ⓒ 1336	Heater-cathode leakage		---	---	Ihk	---	20	µAdc
1256	Electrode current (2) (anode)	Test condition (2)	---	---	Ib	1.3	2.6	mAdc
1256	Electrode current (1) (screen)	Test condition (1)	---	---	Ic2	1.6	3.4	mAdc
1256	Electrode current (2) (screen)	Test condition (2)	---	---	Ic2	0.35	1.0	mAdc
1306	Transconductance (2)	Test condition (1) Ec1 = -35 Vdc	---	---	Sm	1	30	µmhos
1306	Transconductance (3)	Test condition (2); Esig = 0.05 V (see note 3)	---	---	Sm	950	1,450	µmhos
1331	Direct-interelectrode capacitance	Without shield	---	---	Cg1p	---	0.0035	pF
		Without shield			Cin	3.8	6.2	pF
		Without shield			Cout	5.2	8.8	pF
Ⓒ 1101	Secureness of base, cap, or insert		---	---	---	---	---	---
Ⓒ 1211	Insulation of electrodes		---	---	---	---	---	---
Ⓒ 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; Ehk = 100 V	---	---	---	---	---	---
---	Life-test end point (500 hours)	Transconductance (1)	---	---	Sm	1,300	---	µmhos

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.
3. Signal coupled to grid through 1 μF capacitor.
4. Tube types 6SK7GTY and 12SK7GTY are hereby deleted from this specification sheet. For replacement purposes, use tube type 6SK7WA for tube type 6SK7GTY and 12SK7Y for tube type 12SK7GTY.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2401-66)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 83 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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